FabLab / Innovation Center ��� Shahpur University Planning

FabLab / Innovation Center — Design, Equipment & Network-Membership Proposal

For: A new university at Shahpur, India **Prepared by:** Vishvajeet N (drawing on hands-on experience at eirLab, Talence) **Reference models:** eirLab (ENSEIRB-MATMECA, Talence/Bordeaux) · CFI — Center For Innovation (IIT Madras) · MIT / Fab Foundation FabLab standard **Date:** June 2026 · **Currency:** INR (USD reference where useful) **Scope of costing:** equipment + operating costs only. **Building / civil / fit-out costs are deliberately excluded** (assumed provided by the university as space + utilities).

> One-line vision: *A student-run, open-access workshop where any student — regardless of branch or year — can walk in, learn to use a tool safely, and build a physical thing: a circuit, a robot, a 3D print, a laser-cut part, a product prototype.*

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1. Guiding principles

These are the things that make CFI and eirLab work, distilled.

1. **Open access, not a locked teaching lab.** Long opening hours (ideally near-24/7 with student staff), no requirement to be enrolled in a specific course. This is the single biggest differentiator from a normal department lab — and, as Section 8 shows, it's also a hard requirement for joining the Fab Lab network.
2. **Student-run, faculty-backed.** A small core student team runs day-to-day operations (inventory, inductions, machine maintenance). One or two faculty mentors + one full-time technical staff member provide continuity. CFI is almost entirely student-governed; that ownership is *why* it thrives.
3. **Safety as the gate, not bureaucracy as the gate.** Access to a machine = pass a short safety induction for that machine. Everything else stays low-friction.
4. **Tiered tooling.** Cheap, abundant, beginner-safe tools (FDM printers, soldering, hand tools) in the open. Expensive/dangerous tools (laser, CNC, table saw) behind induction + supervision.
5. **Start lean, design for growth.** Phase the equipment buy. A working lab students love beats an empty showpiece.
6. **Project culture over inventory.** Budget for competitions (Robocon/ABU, Formula Student, e-Yantra, hackathons), club projects, and consumables — not just one-time machines.

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2. Reference models — what to borrow from each

Model	What it is	Borrow this
eirLab (Talence)	University FabLab at ENSEIRB-MATMECA, part of the global Fab Foundation network	Fab Foundation equipment standard, induction/badge system, open membership, digital fabrication focus
CFI (IIT Madras)	Large student-run innovation center with themed clubs (Aero, Robotics, 3D printing, Electronics, etc.)	Student governance, club structure, competition teams, 24/7 access culture
MIT / Fab Foundation	The original "Fab Lab" inventory spec + global network	A globally recognised, vendor-neutral equipment checklist; membership in the international network (Section 8)

Recommendation: Adopt the **Fab Foundation equipment baseline** (so you can call it a real "FabLab" and join the network), but organise the *people* and *culture* the **CFI way** (student clubs + competition teams + open access).

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3. Space requirements (university to provide)

Space and civil works are assumed to be provided by the university; no building cost is included here. What the lab *needs* from the space:

3.1 Size

Tier	Usable area	Concurrent users

Starter	~200 m^2 (~2,150 ft^2)	30–40
Standard FabLab (recommended target)	~500 m^2 (~5,400 ft^2)	60–90
Flagship (CFI-scale)	1,000–1,500 m^2	150+

A single large hall plus 2–3 partitionable rooms is enough to start. Ground-floor access (for moving heavy machines/stock) is strongly preferred.

3.2 Zoning (noise, dust, fumes, and safety segregation)

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┌──────────────────────────────────────────────────────────────┐
│  ENTRANCE / RECEPTION + SAFETY INDUCTION DESK + LOCKERS      │
├───────────────┬────────────────────────┬─────────────────────┤
│ DRY ZONE      │  CLEAN DIGITAL FAB     │  ELECTRONICS &      │
│ (quiet)       │  (3D print farm,       │  ROBOTICS LAB       │
│ CAD machines, │   laser cutter w/      │  benches, soldering │
│ meeting/      │   extraction,          │  scopes, PSUs,      │
│ teaching nook │   vinyl cutter)        │  component store    │
├───────────────┴────────────────────────┴─────────────────────┤
│  WET / DIRTY ZONE  (separate ventilation, dust extraction)   │
│  CNC router · woodworking · metal/bench tools · welding bay  │
│  · spray/paint corner · grinder                              │
├────────────────────────────────────────────────────────────┤
│  STORAGE: filament/material store · tool crib · finished     │
│  project shelving · hazmat cabinet                           │
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3.3 Utilities the space must supply (no cost charged to lab budget here)

- **3-phase power** + distributed sockets, RCD/ELCB protection, separate circuits for high-draw machines (laser, CNC, welder).
- **Fume & dust extraction routing** (the extraction *equipment* is in the lab budget, Section 4.5; the ducting/openings to outdoors are a building item).
- **Ventilation & AC** for the dry/electronics/digital zones; the wet zone can be naturally ventilated.
- **Compressed air** point for the dirty zone.
- **Fire-safety provisioning**, marked exits, water point for eyewash.
- **ESD-safe flooring** in the electronics zone; epoxy/concrete in the dirty zone.
- **Wide door / roll-up shutter** to the dirty zone.

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4. Equipment list — by zone (India 2026 price estimates)

Prices are realistic street estimates for India in 2026 and will vary with brand, GST, and education discounts. "Qty" reflects the **Standard tier**. This list is built to **satisfy the Fab Foundation common-inventory requirement** (Section 8).

4.1 Digital fabrication (the FabLab core)

Equipment	Qty	Unit (₹)	Subtotal (₹)	Notes
FDM 3D printers (Bambu/Creality class)	6	40,000	2,40,000	The workhorse "print farm". Quantity > fanciness.
Resin (SLA/MSLA) printer + wash/cure	1	60,000	60,000	Fine detail; needs ventilation + PPE.
Large-format / engineering FDM	1	4,00,000	4,00,000	Bigger build volume, engineering filaments.
CO_2 laser cutter/engraver, 60–100 W + chiller + exhaust	1	5,00,000	5,00,000	Acrylic/wood/MDF. Bed ~600×900 mm+. The most-loved machine in any fablab.
CNC router (desktop→benchtop)	1	2,50,000	2,50,000	Wood/plastic/soft metal.
Vinyl cutter / plotter	1	50,000	50,000	Stickers, stencils, flexible-circuit/copper.
Subtotal			**15,00,000**	

4.2 Electronics & robotics

Equipment	Qty	Unit (₹)	Subtotal (₹)	Notes
ESD soldering/rework stations	8	12,000	96,000	Hot-air rework on a couple.
Digital oscilloscopes (100 MHz)	3	55,000	1,65,000	
Bench power supplies (triple-output)	6	12,000	72,000	
Function generators + bench DMMs	set	—	1,20,000	

Logic analysers / LCR / spectrum (entry)	set	—	80,000	
Reflow oven + stencil setup	1	35,000	35,000	SMD assembly.
Desktop PCB mill *(optional)*	1	3,00,000	3,00,000	Or outsource PCBs and skip this in Phase 1.
Microcontroller/SBC pool (Arduino, ESP32, RPi, **Jetson Nano/Orin**, STM32)	pool	—	2,00,000	Loaner kits + a few AI/vision boards.
Robotics kits, motors, drivers, sensors, LiDAR, IMUs, batteries/chargers	pool	—	3,00,000	Feeds Robocon/competition teams.
Component inventory (R/L/C, ICs, connectors, wire) + storage	—	—	1,50,000	Treat as partly consumable (Section 6).
Subtotal			**15,18,000**	

4.3 Workshop / dirty zone

Equipment	Qty	Unit (₹)	Subtotal (₹)	Notes
Workbenches + vises (heavy)	4	25,000	1,00,000	
Drill press, band saw, table saw, mitre saw	set	—	2,50,000	Induction-gated.
Hand & power tools (drills, drivers, Dremel, etc.)	set	—	1,50,000	
Bench grinder, belt/disc sander	set	—	60,000	
MIG/TIG or arc welder + safety gear	1	1,20,000	1,20,000	Phase 2 optional; serious ventilation needed.
Sewing/textile machine + heat press (Fab standard)	1	45,000	45,000	Soft fabrication, wearables, drone covers.
Subtotal			**7,25,000**	

4.4 Computing, design & metrology

Equipment	Qty	Unit (₹)	Subtotal (₹)	Notes
CAD/CAM workstations (GPU, for CAD + slicing + sim)	6	1,20,000	7,20,000	
Large monitor + projector for teaching nook	—	—	80,000	
Calipers, micrometers, multimeters (qty), bench tools	—	—	60,000	
3D scanner *(optional)*	1	1,50,000	1,50,000	Reverse engineering.
Subtotal (capital)			**10,10,000**	Software is OpEx (Section 6); lean on FOSS — FreeCAD, KiCad, Inkscape.

4.5 Safety & shared equipment

Item	Subtotal (₹)	Notes
Fume extraction arms, dust collector, laser exhaust blower	4,00,000	Often under-budgeted — don't.
PPE: goggles, gloves, aprons, ear/lung protection, face shields	60,000	
Fire extinguishers, smoke/heat detectors, eyewash, first aid	1,00,000	
Signage, induction materials, machine SOP cards, labelling	40,000	
Access control / RFID badge system (ties to induction)	1,50,000	The eirLab-style "earn the badge" model.
Subtotal	**7,50,000**	

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5. Equipment cost summary (Standard tier — equipment only)

Block	Cost (₹)
4.1 Digital fabrication	15,00,000
4.2 Electronics & robotics	15,18,000
4.3 Workshop / dirty zone	7,25,000
4.4 Computing & metrology	10,10,000
4.5 Safety & shared equipment	7,50,000
Equipment subtotal	**55,03,000**
Contingency (~10%)	~5,50,000

TOTAL equipment capital	**≈ ₹60.5 lakh** (~US$71k)

Tier comparison (equipment only):

Tier	Equipment capital (₹)	USD approx
Starter	25–35 lakh	$30–41k
Standard (recommended)	**55–65 lakh**	**$65–76k**
Flagship / CFI-scale	1.5–3 cr	$176–353k

(Building, civil, electrical, HVAC and furniture are excluded by request and are to be provided by the university.)

5a. Benchmarked against the official Fab Foundation equipment standard

The Fab Foundation publishes canonical "what does a Fab Lab cost" figures for **machines + consumables only** (source: Getting Started with Fab Labs, referenced from Growing the Fab Lab Network). At ~₹85/US$ (2026):

Fab Foundation official figure	USD	INR (≈ ₹85/$)
Capital equipment (MIT inventory list)	$25,000 – $65,000	₹21 – 55 lakh
Initial consumables	$15,000 – $40,000	₹13 – 34 lakh
Subtotal to stand up a "regular" Fab Lab	**$40,000 – $105,000**	**₹34 – 89 lakh**
Mobile Fab Lab (reference only)	$298,000	≈ ₹2.5 crore

(UK chapter quotes the equivalent £30–80k capital + £10–25k consumables.)

How this proposal maps onto the benchmark: the Standard equipment line above (**≈ ₹55–60 lakh**) sits right at the **top of the official Fab Foundation capital range** (₹21–55 lakh). It runs to the upper bound because it adds robotics/AI boards, a 6-printer farm, and dirty-zone power tools that the bare inventory omits. Because building/fit-out is excluded here, **this proposal's headline number is now essentially the same as the official Fab Foundation equipment figure** — i.e. we are budgeting a genuine, by-the-book Fab Lab.

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6. Operating cost (annual, Standard tier)

Item	Annual (₹)
Full-time technical staff (1–2: lab engineer + technician)	8–14 lakh
Consumables (filament, sheets, resin, electronic components, blades, bits, PPE refills)	6–10 lakh
Maintenance & spares (laser tube, nozzles, belts, AMC)	3–5 lakh
Software licences (minimise via FOSS)	1–2 lakh
Power (high-draw machines)	3–6 lakh
Competitions, project grants, hackathons, travel	5–15 lakh
Fab network participation (membership is free; Fab Academy tuition optional — see §8)	0–8 lakh
Total OpEx	**≈ ₹26–60 lakh / year**

Sustainability levers: small per-print/material charge-back at cost; sponsored club teams; industry-sponsored machines (named bays); paid weekend workshops for the public/schools (also satisfies the public-access requirement, §8); consultancy/prototyping for local startups (eirLab and many FabLabs do this).

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7. Phasing plan (equipment)

Phase 0 — Pre-launch (Months 0–3) University allocates space + utilities; recruit 1 faculty mentor + 1 technical staff + a 6–8 student core team; draft safety SOPs and the induction/badge system; finalise Phase-1 procurement; **publicly endorse the Fab Charter** (§8).

Phase 1 — "Get students building" (Months 3–8) · equipment ≈ ₹28–35 lakh Commission: FDM print farm (6), 1 laser cutter (+extraction), electronics benches (8), CAD machines (4–6), hand tools, microcontroller/robotics pool, full safety kit. **Open the doors** with public-access hours. This alone is a complete, lovable makerspace and already meets the common-tools criterion.

Phase 2 — "Full FabLab" (Months 8–18) · +equipment ≈ ₹20–28 lakh Add CNC router, large-format + resin printing, dirty-zone power tools, metrology, PCB mill/reflow, vinyl cutter, sewing/textile. **Register on fablabs.io and join the Fab Foundation network** (§8). Launch competition teams (Robocon, e-Yantra, Formula Student feeder).

Phase 3 — "Scale & specialise (CFI-style)" (Year 2+) Welding/metal bay, dedicated club rooms, industry-sponsored bays, public/school outreach, startup incubation tie-in, optional Fab Academy node.

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8. Joining the global Fab Lab network (Fab Foundation) — requirements & application procedure

Becoming a recognised Fab Lab is **free to register** — the "cost" is meeting the criteria below (the equipment in §4 and open-access hours in §1 already do most of the work). Membership gives you the right to use the **FabLab name and logo** (useful for fundraising), a pin on the **world map**, access to the **Fab Academy** programme, and participation in the global community.

8.1 The four requirements (from the Fab Charter)

A space qualifies as a Fab Lab when it meets **all four**:

1. **Public access.** The lab must be open to the public — free or on an in-kind/barter basis — **at least part of the time each week**. (Our open-access model in §1 satisfies this; formalise specific public hours.)
2. **Subscribe to the Fab Charter.** Operate by, and publicly endorse, the Fab Charter — its statements on mission, access, responsibilities, secrecy (designs/processes are shared), and business.
3. **Share the common tool set & processes.** Use broadly the same core machines and software as other Fab Labs (the MIT inventory) so that a project made in one lab can be reproduced in another. Our §4 list is built to this standard.
4. **Participate in the Fab Lab network.** Engage with the wider community — global video conferences, the annual **Fab** conference, knowledge-sharing, and (optionally) hosting/sending students to **Fab Academy**.

8.2 Step-by-step application procedure

1. **Build to the four criteria.** Confirm the §4 inventory covers the common tool set, and commit to weekly public-access hours.
2. **Endorse the Fab Charter** publicly (web page / signage in the lab).
3. **Create a personal account** on the network registry at **fablabs.io**.
4. **Register the lab** at **fablabs.io/labs/new** — add location, photos, equipment, contact, and opening hours. The lab appears on the world map.
5. **Set status `planned` → `active`** once the lab is operational and meeting public-access hours. Submissions pass through a community/regional **approval** step.
6. **Connect with the regional network.** For India/South Asia, engage with the existing Indian Fab Lab community (e.g. **Vigyan Ashram, Pabal** — Asia's first Fab Lab) and the regional Fab coordination for endorsement and mentoring.
7. **Plug into Fab Academy (optional but recommended).** Enrol your technical staff / lead students in **Fab Academy** ("How To Make (Almost) Anything"), or apply to become a **local node** that hosts students. This is the strongest signal of network participation and trains your team on the standard toolchain. *(Budget note: Fab Academy tuition has historically been on the order of US$5k/student/year; hosting as a node spreads this — reflected as the optional ₹0–8 lakh line in §6.)*
8. **Use the logo** in fundraising and signage once you meet all criteria and are "in synchrony with the Fab Lab form and spirit," as the Fab Foundation phrases it.

8.3 What it costs (network side)

Item	Cost
Registering on fablabs.io / world-map listing	**Free**
Using the FabLab name & logo (once compliant)	Free
Meeting the common-inventory requirement	Already in §4 equipment budget
Committing to public-access hours	Operational policy, no capital cost
Fab Academy (optional, per student or as a node)	~US$5k/student/yr historically — optional

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9. People & governance

- **Faculty mentor(s):** 1–2, light-touch, unblock funding/space, sign off on safety policy.
- **Lab engineer / manager (full-time):** keeps machines alive, owns safety, trains the student core.
- **Student core team (8–15):** elected/selected; portfolios = Operations, Inventory, Safety/Inductions, each major machine, Outreach, Finance. *This is the CFI secret sauce.*

- **Member students:** anyone who passes the general safety induction; per-machine badges unlock the gated tools.
- **Clubs/teams under the umbrella:** Robotics, Aero/Drones, 3D-printing, Electronics/IoT, Auto, Software/AI — each owning projects and competition entries.

Access model (eirLab-style): `General induction → member badge` → `per-machine induction → machine badge` → `RFID access during open hours`. Dangerous machines additionally require a trained student/staff "buddy" present.

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10. Risks & how to avoid the common failure modes

Risk	Mitigation
Becomes a locked "showpiece" lab nobody uses	Open access + student governance from day one; measure footfall, not inventory (also a §8 requirement)
Extraction/ventilation skipped → laser/solder fumes, dust	Keep extraction equipment in the lab budget (§4.5); ensure the space provides ducting
Machines break and sit idle	AMC + spares budget + trained student maintainers; buy *fewer, repairable, well-supported* brands
No consumables budget → machines unusable	Ring-fence annual OpEx; introduce at-cost material charge-back
Faculty over-control kills student ownership	Mentor, don't manage; let the student core run operations
Safety incident shuts it all down	Induction gating, PPE, SOP cards, supervision on dangerous tools, insurance

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11. One-paragraph executive summary (for the proposal cover)

> We propose an **open-access, student-run FabLab & Innovation Center** at the new Shahpur campus, modelled on eirLab (Talence) for its digital-fabrication discipline and IIT Madras's CFI for its student-governed, competition-driven culture, and built to the **MIT/Fab Foundation common-inventory standard** so it can **join the global Fab Lab network**. Using university-provided space, the **equipment outlay is ≈ ₹55–65 lakh** (~US$65–76k; phased, with a complete Phase-1 makerspace for ≈ ₹30 lakh), delivering 3D printing, laser/CNC cutting, electronics, robotics and a workshop accessible to **every student** via a simple safety-induction badge system. **Annual operating cost ≈ ₹26–60 lakh.** Network registration on fablabs.io is **free** once the four Fab Charter criteria are met. The center becomes the campus's hands-on engine for tinkering, prototyping, competitions, and student startups — and a recognised node on the worldwide Fab Lab map.

Notes: All costs are 2026 indicative estimates for India (INR), exclusive/inclusive of GST as marked by vendors, and should be validated with 2–3 quotations per major machine before budgeting. USD conversions at ~₹85/$. Building, civil, electrical, HVAC and furniture costs are intentionally excluded.

Sources: Fab Foundation — Getting Started · The Fab Charter (MIT CBA) · fablabs.io registry · Fab Foundation — global community · Fab Foundation UK — how to set up a FabLab